

NEELAY RANJAN

Generative Modeling for Safety-Critical, Data-Scarce Domains

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Generative-modeling researcher building diffusion systems for safety-critical, data-scarce domains. First author on work showing diffusion priors beat SAM, the industry-standard segmentation foundation model, on vascular imaging with under 25 labeled images, at a fraction of the compute. Currently applying the same data-efficient generative approach to real-time aircraft hazard-avoidance routing at NASA. Focus on modular, interpretable pipelines, from segmentation to on-device inference at milliwatt power budgets.

RESEARCH & WORK EXPERIENCE

NASA Ames Research Center

Moffett Field / Mountain View, CA
May 2026 – Present

Generative Modeling Research Intern

- Building weather- and hazard-aware aircraft routing with diffusion models on FAA radar-track (TRX) data, generating flyable trajectories **within ~10 nm (Fréchet) of filed routes** that avoid special-use airspace and launch-related hazard zones for rapid emergency rerouting.
- Implemented a mid-sampling gradient-guidance step that steers the diffusion trajectory around airspace hazards at inference time, enabling hazard rerouting **with no retraining and no new data**; integrated the pipeline into air traffic management (ATM) simulation software used by NASA & FAA.
- Designed a custom LLM with a flight-plan tokenizer that “speaks” filed flight plans, synthesizing a full day of **~44,000 FAA-managed flights** matched to the density and frequency of historical traffic for airspace capacity and safety studies.

Regenstrief Institute

Applied Vision-AI Researcher

Indianapolis, IN
Feb 2024 – Present

- Showed an x0 flipped-objective diffusion segmenter outperforms SAM (ViT-B) on benchmark angiogram segmentation in ultra-low-data regimes: **88.2% Dice with just 16 labels**; used its pre-labels to bootstrap surgeon labeling campaigns, **cutting correction time by 75%**.
- Built a modular SAM-x0 tiled refinement pipeline (SAM prior + partial-resolution x0 diffusion refiner) for real-world, region-agnostic angiograms at native 1024 resolution: **70% Dice with 22 labeled images**, beating both SAM (46%) and raw x0 (32%), while matching full-resolution accuracy at a fraction of the compute.
- Extended the pipeline to cerebral moyamoya MRA, segmenting vasculature from **only 7 labeled images** via patient-level cross-validation and aggressive tiling augmentation: ~55% Dice vs. SAM’s ~20%.

SELECTED PUBLICATIONS

- N. Ranjan et al. “Bootstrapping surgeon labeling campaigns with x0-diffusion: label-efficient vessel segmentation of catheter-based angiograms.” *First author, in preparation.*
- N. Ranjan et al. “Modular SAM-prior diffusion refinement for real-world angiogram and seven-shot moyamoya MRA vessel segmentation.” *First author, in preparation.*
- Co-author, “Helical Antenna for Electromagnetic Field Stimulation in Alzheimer’s Disease Therapy.” *Presented at IEEE MWSCAS 2026; led the PCB design team.*

PROJECTS & APPLIED RESEARCH

Energy-Based Model for Chess → Solar-Powered Inference Device

2025

- Trained an energy-based model that learns an energy landscape over game states to score & rank moves, reaching **~1900–2200 estimated Elo** against a ladder of Stockfish depths with known Elo ratings.
- Ran quantization experiments and found FP32 outperformed INT8 on ARM (no native INT8 kernels); embedded the EBM on a Raspberry Pi Zero 2 W (~1 s / move) as a standalone interactive chess engine **powered solely by onboard solar & supercapacitors**.

LLM Design Team Lead – Aircraft Maintenance Assistant for V2X (Raytheon divestiture)

Aug 2024 – May 2025

- Led design and fine-tuning of a maintenance-assistant LLM whose two-stage RAG pipeline (summary-level source selection → section-level pinpointing with direct quoting) **cut hallucinations from ~40% to ~5%**.

KEY RESULTS

-  **~10 nm Fréchet**
Hazard-avoiding routes, no retraining
-  **~44,000 flights**
One synthetic day of US traffic via LLM
-  **88.2% Dice at 16 labels**
x0 segmenter vs. SAM on angiograms
-  **75% faster correction**
x0 pre-labels speed surgeon labeling
-  **70% Dice at 1024 px**
Real-world angiograms (SAM 46%)
-  **7 training images**
Moyamoya MRA at ~55% (SAM ~20%)
-  **40% → 5% hallucinations**
Two-stage RAG, aircraft maintenance LLM

EDUCATION

Purdue University, College of Engineering

B.S. in Artificial Intelligence

Expected May 2027 Indianapolis, IN
Concentration: Intelligent Control & Systems
Minor: Mathematics
GPA: 3.68 / 4.0 · Dean’s List
John Martinson Honors College
CITI: Biomedical Research, RCR, GCP

TECHNICAL SKILLS

Generative modeling

- diffusion (x0 / v-prediction)
- energy-based models
- foundation-model adaptation (SAM)
- transformers LLM fine-tuning & RAG
- safety-critical AI interpretability
- multimodal pipelines

ML engineering

- Python C/C++ CUDA PyTorch
- transfer learning
- medical image segmentation quantization
- Docker Slurm / H100 W&B
- Linux / cluster administration Git

Hardware / edge

- PCB design ultra-low-power & solar inference
- embedded ML